Chapter 3

ARITHMETIC SEQUENCE AND ALGEBRA

Section A

1. The sum of the first n-terms of an arithmetic sequence is $3n^2+6n$, , what is the algebraic form of the sequence?

   A) 6n+3 B) 6n-3

   C) 3n+3 D) 3n-3

2. Statement 1 : If the algebraic form of an arithmetic sequence is 6n+3, then the sum of the first 20 terms of the sequence is 1320.

   Statement 2 : If the algebraic form of the arithmetic sequence is an+b then the sum of the first 'n' terms is $\frac{an(n+1)}{2}+bn$.

   A) Statement 1 is correct, statement 2 is wrong

   B) Statement 1 is wrong, statement 2 is correct

   C) Both the statements are true, statement 2 is the correct explanation of statement 1

   D) Both the statements are true, statement 2 is not the correct explanation of statement 1

3. Statement 1 : If the algebraic form of the sum of an arithmetic sequence is $2n^2+n$, then the algebraic form of the sequence is $x_n = 4n+1$.

   Statement 2 : The algebraic form of the sum of an arithmetic sequence with $x_n = an+b$ is Pn^2+qn where $P=\frac{a}{2}$, $q=\frac{a}{2}+b$.

   A) Statement 1 is correct, statement 2 is wrong

   B) Statement 1 is wrong, statement 2 is correct

C) Both the statements are true, statement 2 is the correct explanation of statement 1

D) Both the statements are true, statement 2 is not the correct explanation of statement 1

4. Statement 1 : The sum of the first 20 terms of the arithmetic sequence 3, 6, 9, ... is 630.

   Statement 2 : The sum of first n multiples of 3 is $\frac{3n(n+1)}{2}$

A) Statement 1 is correct, statement 2 is wrong

B) Statement 1 is wrong, statement 2 is correct

C) Both the statements are true, statement 2 is the correct explanation of statement 1

D) Both the statements are true, statement 2 is not the correct explanation of statement 1

Section B

1. 63, 65, 67, ..., and 3, 10, 17, ... are two arthmetic sequences. Can any term the terms of these sequences in the same position are equal? If so write the term and position?

2. In the arithmetic sequence 10, 16, 22, ... Can the sum of any two terms of this sequence be a term of this same sequence?

3. How many three digit numbers are there, which are multiples of 7?

4. If 17th term of an arithmetic sequence is 7 more than the 10th term? Find the common difference of this sequence?

5. The sum of first 9 terms of an arithmetic sequence is 261, sum of next 6 terms is 444.

   a) Write the 5th term

b) Write the 8th term

c) Write the algebriac form of the sequence

d) Write the algebriac form of the sum of the sequence.

6. Find the following sums.

i) $1 + 2 + 3 + \ldots\ldots\ldots\ldots + 20$

ii) $4 + 8 + 12 + \ldots\ldots\ldots\ldots + 80$

iii) $7 + 11 + 15 + \ldots\ldots\ldots\ldots 83$

iv) $10 + 17 + 24 + \ldots\ldots\ldots\ldots + 143$.

7. The first term of an arithmetic sequence is 8, the common difference of this sequence is 4.

i) Write the algebriac form of the sequence.

ii) Write the algebriac form of the sum of the sequence.

8. The algebriac form of the sum of an arithmetic sequence is $2n^2 + 4n$.

i) Find the first term of this sequence

ii) Find the common difference of this sequence

iii) How many terms of this sequence starting from the first is to be added to get the sum 240?

9. Consider the arithmetic sequence 3, 8, 13, 18, ...

i) What is remainder obtained when, the terms of this sequence are divided by 5?

ii) Is 503 a term of this sequence?

iii) Write the algebraic form of this sequence?

10. i) Write the algebraic form of the arithmetic sequence 6, 10, 14, ...

ii) Write the sequence of natural numbers among the terms of the arithmetic seuqnce $\frac{6}{7}, \frac{10}{7}, 2, \ldots\ldots\ldots\ldots\ldots?$

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Chapter 3

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ANSWERS

Section A

1. A
2. C) Both the statements are true, statement 2 is the correct explanation of statement 1
3. B) Statement 1 is wrong, statement 2 is correct
4. C) Both the statements are true, statement 2 is the correct explanation of statement 1

Section B

1. 63, 65, 67, ...

   $x_n = 2n+61$

   3, 10, 17, ... $x_n = 7n - 4$

   In n^{th} term of two sequences are equal,

   $2n + 61 = 7n - 4$

   $5n = 65$

   $n = \frac{65}{5}$

   is n = 13

   In both sequences 13th terms are equal

   13th term = $3+(12x7)$

   = 3 + 84 = 87

2. Algebra form

$x_n = 6n + 4$

Each term of sequence is divided by 6, remainder is 4.

n^{th} term $x_n = 6n + 4$

n^{th} term $x_m = 6m + 4$

$x_n + x_m = 6(n+m) + 6 + 2$

ie, $x_n + x_m$ is divided by 6, remainder is 2.

3. First 3 digit number which is multiple of 7 = 105

Last 3 digit number which is multiple 7 = 994

$\therefore 105 + (n-1) \times 7 = 994$

$105 + 7n - 7 = 994$

$7n = 896$

$n = \frac{896}{7} = 128$

128 three digit number which is divisible by 7.

4. **d = 1**

5. a) 5th term = $\frac{261}{9} = 29$

b) Sum of 15th terms = $261 + 444 = 705$

8th term = $\frac{705}{15} = 47$

$3d = x_8 - x_5 = 47 - 29 = 18$

$d = \frac{18}{3} = 6$

$x_1 = x_5 - 4d$

$= 29 - 24 = 5$

c) 6n-1

d) $3n^2 + 2n$

6. i) 210

ii) 840

iii) 900

iv) 900 + 840 - 210 = 1530

7. i) 4n+4

ii) $\frac{4n(n+1)}{2} + 4n = 2n^2 + 6n$

8. i) 6

ii) 4

iii) $2n^2+4n = 240$

$n^2 +2n = 120$

$(n+1)^2 = 121$

n+1 = 11

n=10

Number of terms = 10

9. i) 3

ii) Yes

iii) 5n-2

10. i) 4n+2

ii) $x_n = \frac{4n+2}{7}$

Substituting n = 3, 10, 17, ... get the perfect number

2, 6, 10, ...